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UniCare: Interoperable Electronic Health Record (EHR) Platform

24MCAR295 - Software Project Part 1

ABSTRACT

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ABSTRACT

UniCare: Interoperable Electronic Health Record (EHR) Platform

A unified digital healthcare platform that securely maintains a patient's longitudinal health record across participating healthcare providers.

Problem Statement

Healthcare institutions typically maintain patient records independently, resulting in fragmented medical histories that cannot be easily accessed across different healthcare providers. Patients are often required to carry physical prescriptions, laboratory reports, and diagnostic scan reports when consulting different hospitals, leading to repeated medical tests, duplicate prescriptions, delayed treatment, and an increased risk of medication errors. During emergency situations, the absence of complete medical history may delay appropriate clinical decisions. Existing Hospital Management Systems (HMS) primarily focus on managing healthcare operations within individual hospitals and are not designed to maintain a unified patient-centric health record across multiple participating healthcare providers.

Proposed Solution

The proposed UniCare: Interoperable Electronic Health Record (EHR) Platform is a centralized, patient-centric healthcare platform that maintains a Unified Patient Health Record (UPHR) using a unique Health ID. Hospitals that wish to become part of the UniCare platform can submit a registration request. After approval by the Super Administrator, the hospital receives a unique Hospital ID and becomes a participating healthcare provider on the platform.

The platform enables patients, doctors, receptionists, hospital administrators, and super administrators to securely access and manage healthcare information using role-based access control and patient consent. Patients are registered by authorized receptionists at participating hospitals and receive a unique Health ID. They can securely access their complete medical history through a dedicated web portal, generate a Digital Health Card containing their unique Health ID, upload laboratory and diagnostic reports obtained from external diagnostic centers,

view digital prescriptions, receive medicine reminders, and book appointments with participating healthcare providers.

Doctors can securely search patients using the Health ID, access authorized patient medical records, review the patient's complete longitudinal health history, record diagnoses, generate digital prescriptions, upload laboratory and diagnostic reports, and determine whether medical records, prescriptions, and laboratory reports should be shared with other participating hospitals. Hospital administrators can manage doctor and receptionist accounts, manage appointment schedules, monitor hospital activities, manage hospital users, and view platform usage statistics. Super administrators manage hospital registration requests, approve participating hospitals, and oversee the overall platform.

The system is developed using React.js, Django, and MySQL.

Project Development

The project focuses on developing the complete interoperable Electronic Health Record platform, including secure authentication, hospital registration and approval, unique Health ID generation, patient registration, Digital Health Card generation, role-based access control, appointment management, digital prescriptions, laboratory and diagnostic report management, patient web portal, doctor and hospital administrator web portals, and controlled sharing of essential patient records across participating healthcare providers.

The platform maintains a Unified Patient Health Record (UPHR) that enables authorized healthcare professionals to securely access a patient's essential medical history across participating hospitals. Doctors can classify medical records, prescriptions, and laboratory reports as either Shared or Local. Only records marked as Shared are accessible to authorized healthcare providers in other participating hospitals, while Local records remain accessible only within the treating hospital. Patients can view all of their own medical records through the patient portal.

Expected Outcome

The proposed UniCare platform improves continuity of patient care by providing authorized healthcare professionals with secure access to a patient's Unified Patient Health Record across participating healthcare providers. It minimizes duplicate investigations, reduces repeated prescriptions, decreases the need for patients to carry physical medical records, improves continuity of care, and enables faster access to essential patient history during emergency situations. By providing secure interoperability and controlled sharing of essential medical information within participating healthcare providers, UniCare contributes to safer, more efficient, coordinated, and patient-centered healthcare delivery while preserving patient privacy and ensuring secure healthcare information exchange.

Target Users and Functionalities

1. Patient

Functionalities

- Login
- Receive a Unique Health ID
- Generate Digital Health Card (E-Health Card)
- View Medical History
- View Digital Prescriptions
- Book Appointments with Participating Healthcare Providers

2. Doctor

Functionalities

- Login
- Search Patient using Unique Health ID
- Access Authorized Patient Medical Records
- View Complete Longitudinal Patient History
- Record Diagnosis
- Generate Digital Prescriptions
- Upload Laboratory and Diagnostic Reports
- View Patient Uploaded Reports



- Mark Medical Records as Shared or Local

3. Hospital Administrator

Functionalities

- Register Healthcare Organization on UniCare
- Register and Manage Doctor Accounts
- Register and Manage Receptionist Accounts
- Manage Departments
- Manage Appointment

4. Receptionist

Functionalities

- Register patients
- Search existing patients
- Book appointments

5. Super Admin

Functionalities

- Login
- Approve Hospital Registration Requests
- Manage Participating Hospitals
- View analytics
- Generate reports
- Generate Hospital ID

Suggested Technologies

Frontend

- React.js (Web Portal)
- HTML5
- CSS3
- Bootstrap



Backend

- Django (Python)

Database

- MySQL

Other Tools

- Git & GitHub
- Visual Studio Code
- Figma

